24V input on GIGA shield | Local 3.3V display conversion | Molex 15018-0563 Type-A 60-way FFC, 102mm nominal

display

Datei: display.kicad_sch

ribbon

Datei: ribbon.kicad_sch

power

Datei: power.kicad_sch

logic_power

Datei: logic_power.kicad_sch

sequencing

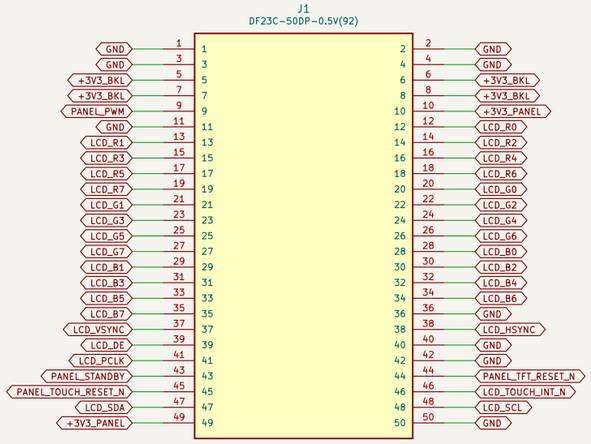
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Supply present

Supply present

ProtoMate		
Sheet: /		
File: display-adapter.kicad_sch		
Title: display-adapter / Smartwin display interface		
Size: A3	Date: 2026-10-22	**Rev: F**
KiCad E.D.A. 10.0.6		Id: 1/6

Bottom-mounted DF23 plug. Pin orientation follows Smartwin mechanical drawing Detail A. All supply/GND pins used.



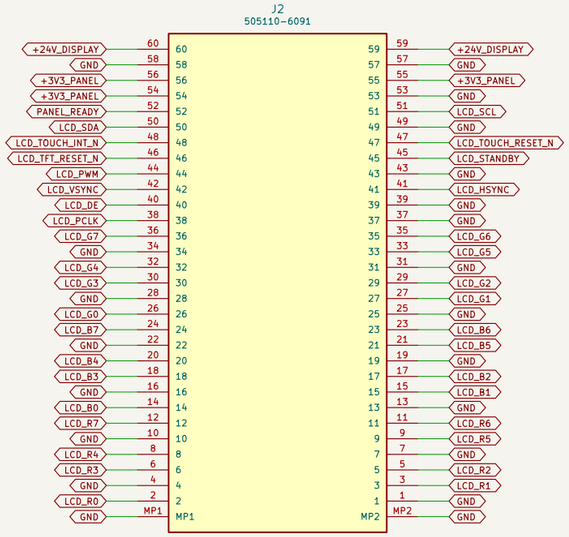
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File: display.kicad_sch

Title: display-adapter / display

Size: A3	Date: 2026-10-22	**Rev: F**
KiCad E.D.A. 10.0.6		Id: 2/6

60-way Molex 15018-0563 Type-A / 505110-6091 sockets; 0.5mm pitch, 0.30mm mating thickness; GIGA n <-> display 61-n.



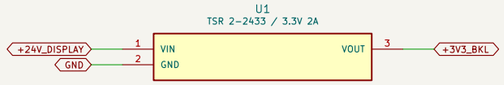
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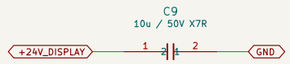
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File: ribbon.kicad_sch

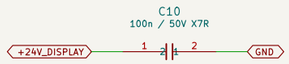
Title: display-adapter / ribbon

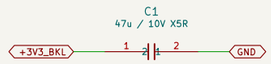
Size: A3	Date: 2026-10-22	**Rev: F**
KiCad E.D.A. 10.0.6		Id: 3/6

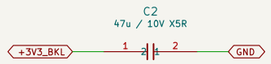
U1: 24V from ribbon –> BACKLIGHT 3.3V / 2A. Never join panel 3.3V to GIGA 3.3V. Module is hand soldered; trim leads.





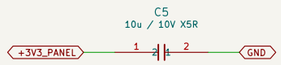


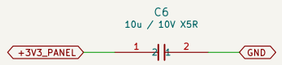




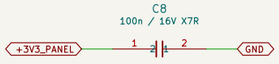


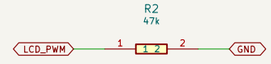


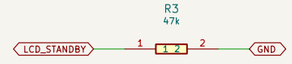




































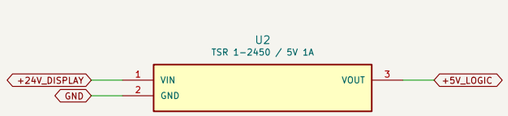
ProtoMate

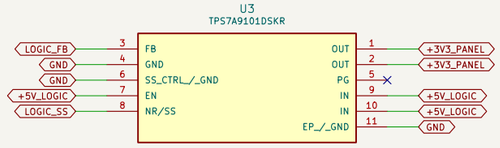
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File: power.kicad_sch

Title: display–adapter / power

Size: A3	Date: 2026-10-22	**Rev: F**
KiCad E.D.A. 10.0.6		Id: 4/6

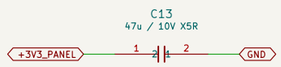
U2: 24V -> 5V. U3: precision 3.296V, 47nF soft start (~6ms). 0.1% feedback resistors. EP must be soldered.

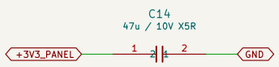


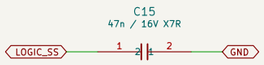




















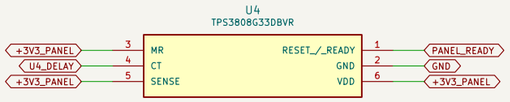
ProtoMate

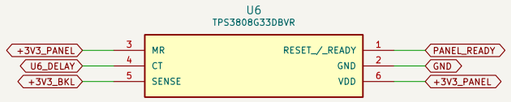
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File: logic_power.kicad_sch

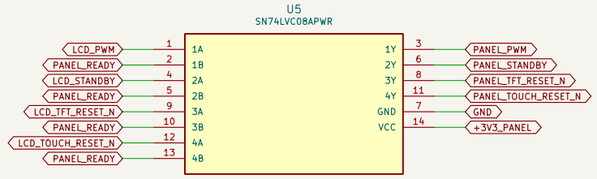
Title: display-adapter / logic_power

Size: A3	Date: 2026-10-22	**Rev: F**
KiCad E.D.A. 10.0.6		Id: 5/6

Both supplies valid for 300ms before READY. Hardware gates reset, standby and PWM. READY returns to GIGA D70.



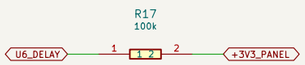


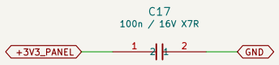


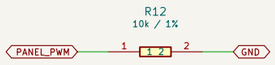




















ProtoMate

Sheet: /sequencing/
File: sequencing.kicad_sch

Title: display–adapter / sequencing

Size: A3	Date: 2026-10-22	**Rev: F**
KiCad E.D.A. 10.0.6		Id: 6/6